

 Section: Particles and Waves

 Topic: Spectra and the Bohr Model

(a) State one other feature of the Bohr model

✔ 1 mark

✔ **Answer:**

The electron can move between energy levels by **absorbing or emitting discrete amounts of energy**.

(b)(i) Explain how a line emission spectrum is produced

✔ 2 marks

✔ **Answer:**

- Electrons in atoms **gain energy** and move to **higher energy levels (excited states)**
- When electrons **fall back down** to lower energy levels, they emit **photons**
- Each transition produces a photon of **specific energy**, resulting in **discrete spectral lines**

(b)(ii) Explain why some lines appear brighter than others

✔ 2 marks

✔ **Answer:**

Brighter lines correspond to **transitions that occur more frequently**, i.e. **more electrons** making that

(c)(i) Number of possible emission lines between 5 energy levels

✔ 1 mark

Use formula:

Number of lines = $\frac{n(n-1)}{2} = \frac{5(5-1)}{2} = 10$

(c)(ii) (A) Calculate the frequency for E₄ → E₁ transition

✔ 3 marks

Given:

- $E_4 = -0.871 \times 10^{-19} \text{ J}$
- $E_1 = -5.45 \times 10^{-19} \text{ J}$

$\Delta E = E_1 - E_4 = (-5.45 - (-0.871)) \times 10^{-19} = 4.579 \times 10^{-19} \text{ J}$

$f = \frac{E}{h} = \frac{4.579 \times 10^{-19}}{6.63 \times 10^{-34}} = 6.91 \times 10^{14} \text{ Hz}$

(c)(ii) (B) State wavelength of blue-green spectral line

✔ 1 mark

✔ **Answer:**

486 nm

(This is provided in the spectral line data sheet)

(c)(ii) (C) Determine the observed wavelength due to redshift

✔ 5 marks

Use redshift formula:

$\lambda_{\text{obs}} = \lambda_{\text{rest}} \left(1 + \frac{v}{c} \right)$
 $= 486 \times \left(1 + \frac{4.52 \times 10^6}{3.00 \times 10^8} \right) = 486 \times (1 + 0.01507)$
 $= 493 \text{ nm}$

 Revision Tips

- **Bohr model:** discrete orbits, energy levels, photon emission
- Number of lines = transitions between levels: use $n(n-1)/2$
- **Energy → frequency:** $f = \frac{E}{h}$
- **Redshift** increases wavelength when objects move away